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1 Metric Space Essentials

1.1 Compactness

Theorem 1 (Bolzano-Weierstrass). *Every bounded sequence in $\mathbb{R}$ has a convergent subsequence.*

Proof. Let $\{x_n\}_{n \in \mathbb{N}}$ be a bounded sequence in $\mathbb{R}$, so that there are values $m \leq M$ so that

$$m \leq x_n \leq M \quad \forall n \in \mathbb{N}.$$

We will construct a subsequence $\{x_{n_i}\}_{i \in \mathbb{N}} \subset \{x_n\}_{n \in \mathbb{N}}$, choosing terms so that the result is a Cauchy sequence. Because $\mathbb{R}$ is a complete metric space, the subsequence converges. To start, we will pick the first element. Divide the interval $[m, M]$ into a number N_0 of subintervals,

$$\begin{aligned} &[m, m+1], \\ &[m+1, m+2], \\ &\dots, \\ &[m+N_0-1, m+N_0], \end{aligned}$$

so that $m+N_0 \geq M$. Because there are finitely many intervals and infinitely many terms in the sequence, there is at least one interval I_0 with infinitely many terms. Pick x_{n_0} to be one element of the sequence belonging to I_0 .

We will now show how to construct inductively the i^{th} term of the subsequence, for $i \geq 1$. Divide the interval $I_{i-1} = [m_{i-1}, M_{i-1}]$ into a number N_i of subintervals,

$$\begin{aligned} &[m_{i-1}, m_{i-1} + \frac{1}{i+1}], \\ &[m_{i-1} + \frac{1}{i+1}, m_{i-1} + \frac{2}{i+1}], \\ &\dots, \\ &[m_{i-1} + \frac{N_i-1}{i+1}, m_{i-1} + \frac{N_i}{i+1}]. \end{aligned}$$

Because I_{i-1} contains infinitely many terms of the original sequence, at least one of the above intervals contains infinitely many terms of the original sequence, and we label it I_i . We can also choose an element x_{n_i} so that $x_{n_i} \in I_i$ and $n_i > n_{i-1}$.

With our subsequence in hand, we can see that for any i , any subsequent term of the subsequence resides with x_{n_i} in an interval of width $\frac{1}{i+1}$. So our subsequence is Cauchy and thus also convergent.

□

The above result also holds in $\mathbb{R}^n$, $\mathbb{C}^n$, and other well-behaved spaces. We will see later that in infinite-dimensional vector spaces, boundedness does not automatically give us a compact closure.

Theorem 2. *The normed linear space $(\mathcal{C}(K), \|\cdot\|_\infty)$ of continuous functions on a compact metric space K is complete.*

Proof. Let $\{f_n\}_{n \in \mathbb{N}} \subset \mathcal{C}(K)$ be a Cauchy sequence:

$$\lim_{n, m \rightarrow \infty} \|f_n - f_m\|_\infty = 0.$$

For each $x \in K$, $f_n(x)$ is a Cauchy sequence, and because $\mathbb{R}$ is complete, the limit

$$f(x) := \lim_{n \rightarrow \infty} f_n(x) \text{ exists.}$$

To get our result, we need to show two things. First, we will show that f_n converges to f uniformly, and then we will show that f is continuous.

- To show that f_n converges to f uniformly, let $\epsilon > 0$. Because f_n is Cauchy,

$$\exists N \in \mathbb{N}, \forall n, m \geq N, \forall x \in K, |f_n(x) - f_m(x)| < \epsilon/2.$$

Because f_n converges pointwise to f , we also have

$$\exists M_x > N, \forall m \geq M_x, |f_m(x) - f(x)| < \epsilon/2.$$

By the triangle inequality,

$$\forall x \in K, \forall n \geq N, |f_n(x) - f(x)| < \epsilon.$$

So f_n converges uniformly to f .

- Now we show that f is continuous. Let $x \in K$, and let $\epsilon > 0$. From the uniform convergence just established,

$$\exists N \in \mathbb{N}, \forall n \geq N, \forall x \in K, |f_n(x) - f(x)| < \epsilon/3.$$

Because f_N is continuous,

$$\exists \alpha > 0, \forall y \in K \text{ with } \text{dist}(x, y) < \alpha, |f_N(x) - f_N(y)| < \epsilon/3.$$

By the triangle inequality,

$$\begin{aligned} |f(x) - f(y)| &\leq |f(x) - f_N(x)| + |f_N(x) - f_N(y)| + |f_N(y) - f(y)| \\ &< \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} \\ &= \epsilon. \end{aligned}$$

So f is continuous.

□

If we look at the above proof, none of the steps require that K is compact. We focus on compact domain because it allows us to talk about a “space of continuous functions”. If the domain is not compact, then not all continuous functions are bounded. We need to be able to put a norm on all elements for our space to be well defined. We will cover a lot of results on *Banach spaces*, which are complete normed linear spaces. $\mathcal{C}([0, 1])$ is maybe the prime example for why we study Banach spaces without jumping directly to Hilbert spaces, wherein the norm is induced by an inner product. There is no inner product that induces the infinity norm on $\mathcal{C}([0, 1])$.

Also in the above proof, we established an additional result that is handy to keep around.

Lemma 1. *The uniform limit of a sequence of continuous functions is continuous.*

2 Vector Space Essentials

2.1 Riesz’s lemma

The following lemma is useful when arguments based on orthogonality would be useful, but we don’t have an inner product to work with. In Hilbert space,

you can Gram-Schmidt your way to a bigger basis, and that only works when your norm is induced by an inner product, but Riesz's lemma gives you another way to procure elements from a bigger subspace. A common application is to demonstrate that a set is not compact by building a sequence where no two elements are within distance ϵ .

Here is a statement and proof from Kreyszig [?], with my attempt to highlight the crux of the argument.

Lemma 2 (Riesz's lemma). *Let Y and Z be linear subspaces of a normed space X , with Y a closed, proper subset of Z . Then for all $\theta \in (0, 1)$, there is a $z \in Z$ such that*

$$\|z\| = 1, \quad \|z - y\| \geq \theta$$

for all $y \in Y$.

Proof. Let $\theta \in (0, 1)$. We will rely heavily on the definition of infimum to pull an element that gets within the " θ -scale" of achieving it. Because Y is a proper subset of Z , there is an element $v \in Z \setminus Y$, and because Y is closed,

$$d(v, Y) = \inf_{y \in Y} \|v - y\| = \alpha > 0.$$

Because $0 < \theta < 1$, and using the definition of infimum, there exists an element $y_0 \in Y$ such that

$$\alpha \leq \|v - y_0\| \leq \frac{\alpha}{\theta}.$$

Now we define our element z as

$$z = c(v - y_0), \quad c = \|v - y_0\|^{-1}.$$

Let $y \in Y$. We need to show that $\|z - y\| \geq \theta$.

$$\begin{aligned} \|z - y\| &= \|c(v - y_0) - y\| \\ &= c\|(v - y_0) - c^{-1}y\| \\ &= c\|v - y_1\| \\ &= \|v - y_0\|^{-1} \cdot \|v - y_1\| \\ &\geq \|v - y_0\|^{-1} \cdot \inf_{y \in Y} \|v - y\| \\ &= \|v - y_0\|^{-1} \cdot \alpha, \end{aligned}$$

with $y_1 = (y_0 + c^{-1}y) \in Y$. Critically, we can invert the earlier inequality to obtain

$$\frac{\theta}{\alpha} \leq \|v - y_0\|^{-1} \leq \frac{1}{\alpha},$$

so that

$$\|z - y\| \geq \frac{\theta}{\alpha} \cdot \alpha = \theta.$$

□

An application of Riesz's lemma, the following theorem illustrates the trouble we get into when working in infinite dimensions. It's also frequently useful when determining compactness of sets and operators.

Theorem 3 (Riesz's theorem). *For X a normed space, $\overline{B_X(0, 1)}$ is compact if and only if X has finite dimension.*

Proof.

- Let X have finite dimension N . Then there are unit vectors $\{u_1, \dots, u_N\} \subset X$ with $\|u_i\| = 1$ and $\text{span}\{u_1, \dots, u_N\} = X$. Now consider any sequence

$$\{x_n\}_{n \in \mathbb{N}} \subset \overline{B_X(0, 1)}.$$

Using the basis, for each n ,

$$x_n = a_{n1}u_1 + \dots + a_{nN}u_N,$$

and by the triangle inequality,

$$\|x_n\| \leq \sum_{k=1}^N |a_{nk}|.$$

Because $\{x_n\}_{n \in \mathbb{N}}$ is bounded, so are the coefficients in the above expansion. So the sequence

$$\{(a_{n1}, \dots, a_{nN})\}_{n \in \mathbb{N}} \subset \mathbb{R}^N$$

is bounded and thus has a convergent subsequence by the Bolzano-Weierstrass theorem. Using the triangle inequality as above, the corresponding subsequence of $\{x_n\}_{n \in \mathbb{N}}$ converges as well. Given this outcome, because $\overline{B_X(0, 1)}$ is closed, it is compact.

- Let X have infinite dimension. Let $u_0 \in X$ with $\|u_0\| = 1$. Then $\text{span}\{u_0\}$ is a proper, closed subspace of X . By Riesz's lemma,

$$\exists u_1 \in X \setminus \text{span}\{u_0\}, \|u_1\| = 1, \inf_{v \in \text{span}\{u_0\}} \|u_1 - v\| \geq \frac{1}{2}.$$

By induction, because $\text{span}\{u_0, \dots, u_{n-1}\}$ is a proper subspace of X with finite dimension, it is a closed subspace, and Riesz's lemma again gives

$$u_n \in X \setminus \text{span}\{u_0, \dots, u_{n-1}\}, \|u_n\| = 1, \inf_{v \in \text{span}\{u_0, \dots, u_{n-1}\}} \|v - u_n\| \geq \frac{1}{2}.$$

Because $\|u_n - u_m\| \geq 1/2$ for all $n \neq m$, $\{u_n\}_{n \in \mathbb{N}} \subset \overline{B_X(0, 1)}$ has no convergent subsequence. So $\overline{B_X(0, 1)}$ is not compact.

□

2.2 Linear Maps

Theorem 4. *Let $T : X \rightarrow Y$ be a linear map between normed spaces X and Y . $\ker(T)$ is a linear subspace of X , and $\text{range}(T)$ is a linear subspace of Y . Additionally, if T is bounded, then $\ker(T)$ is a closed subset of X .*

Proof. We will just prove that $\ker(T)$ is closed when T is bounded:

Let $\{x_n\}_{n=1}^\infty$ be a convergent sequence in $\ker(T)$, with $x_n \rightarrow x \in X$. By continuity, we can exchange the limit with T as in:

$$Tx = T\left(\lim_{n \rightarrow \infty} x_n\right) = \lim_{n \rightarrow \infty} Tx_n = \lim_{n \rightarrow \infty} 0 = 0.$$

So $x \in \ker(T)$, and thus $\ker(T)$ is closed.

□

The following theorem is the most essential result in linear functional analysis.

Theorem 5. *A linear map is bounded if and only if it is continuous.*

Proof.

- Let $T : X \rightarrow Y$ be bounded. Then there is a value $K \geq 0$ such that for all $x \in X$, $\|Tx\| \leq K\|x\|$. Let $\epsilon > 0$. Then for any $z \in X$ with $\|x - z\| < K^{-1}\epsilon$,

$$\|Tx - Tz\| \leq K\|x - z\| < \epsilon.$$

So T is continuous.

- Let $T : X \rightarrow Y$ be continuous. Let us consider in particular the continuity of T at $0 \in X$. Let $C > 0$. There is a value α such that for any $z \in X$ with $\|z\| \leq \alpha$, $\|Tz\| \leq C$. Now let $x \in X$. Using the homogeneity of T ,

$$\|Tx\| = \alpha^{-1}\|x\| \cdot \|T(\alpha\|x\|^{-1}x)\| \leq \alpha^{-1}C\|x\|.$$

So T is bounded.

□

3 Banach Space

3.1 Compact Operators

When we seek to show that a particular operator is compact, it is often useful to employ the following result. If you can show that your operator is the uniform limit of a sequence of compact operators, then your operator is compact.

Theorem 6. *If X is a Banach space, $T_n : X \rightarrow X$ is a sequence of compact operators, and T is a linear map, then if*

$$\lim_{n \rightarrow \infty} \|T_n - T\| = 0,$$

then T is a compact operator.

Proof. To prove by contrapositive, suppose that T_n is a sequence of operators that converges uniformly to T , and suppose that T is not compact. Then we can supply a sequence $\{x_k\}_{k \in \mathbb{N}}$, with $\|x_k\| \leq 1$, such that Tx_k does not have a convergent subsequence. We will show that at least one of the operators T_N in the sequence is not compact.

Let $\{x_{k_i}\}_{i \in \mathbb{N}}$ be any subsequence of x_k . Then Tx_{k_i} does not converge, and thus Tx_{k_i} is not Cauchy. So by inverting the definition of convergence, we get

$$\exists \epsilon > 0, \forall i \in \mathbb{N}, \exists I, J \geq n, \|Tx_{k_I} - Tx_{k_J}\| \geq 3\epsilon.$$

Using the uniform convergence of T_n , we can say

$$\exists N \in \mathbb{N}, \|T_N - T\| \leq \epsilon \forall x \in X, \|x\| \leq 1.$$

So then using the previous statement,

$$\|T_N x_{k_I} - T_N x_{k_J}\| \geq \epsilon.$$

So T_N is not compact.

□

3.2 Duals and Adjoints

Proposition 1. *For any Banach space X , the topological dual X' is also a Banach space.*

Proof. Let $\{\varphi_n\}_{n \in \mathbb{N}} \subset X'$ be a Cauchy sequence in the operator norm, $\|\cdot\|_\infty$. Then for any $x \in X$, $\varphi_n(x)$ is a Cauchy sequence in $\mathbb{R}$, which is complete, so let $\varphi(x)$ be the pointwise limit. Because $\{\varphi_n\}_{n \in \mathbb{N}}$ is Cauchy in the uniform norm and converges pointwise to φ , φ_n converges uniformly to φ . Because any uniform limit of a sequence of continuous functions is continuous, φ is continuous. To finish, we show that φ is linear. Let $x, y \in X$ and let $a, b \in \mathbb{R}$. Then

$$\varphi(ax + by) = \lim_{n \rightarrow \infty} \varphi_n(ax + by) = \lim_{n \rightarrow \infty} (a\varphi_n(x) + b\varphi_n(y)) = a\varphi(x) + b\varphi(y).$$

□

In fact, the above result holds if we replace the dual with any space of bounded linear maps into a Banach space. Stated formally, for X a normed space and Y a Banach space, the space $\mathcal{B}(X, Y)$ of bounded linear operators from X into Y is complete with respect to the operator norm $\|\cdot\|_\infty$.

3.3 The Baire Category Theorem

Baire's theorem brings together cardinality and topology, and it's used to prove many of the big results on Banach spaces. It can be used in concrete scenarios if you recognize that you're working with a countable subdivision of a complete space.

Definition 1. *In a metric space (X, d) , $A \subset X$ is nowhere dense if the closure of A has empty interior:*

$$\text{int}(\overline{A}) = \emptyset.$$

Definition 2. *In a metric space (X, d) , $B \subset X$ is meager if B is a countable union of nowhere-dense sets.*

Definition 3. *A space X is a Baire space if any of the following equivalent properties hold:*

1. *Every meager set has empty interior;*
2. *A countable union of closed sets with empty interior has empty interior;*
3. *A countable intersection of everywhere dense open sets is dense.*

Theorem 7 (Baire category). *Every complete metric space is a Baire space.*

3.4 The Open Mapping and Closed Graph Theorems

The open mapping theorem tells us, essentially, that a bounded, invertible linear operator between Banach spaces has a bounded inverse. Invertibility is a stringent requirement, but consider—any operator can be made surjective by taking the restriction of its codomain to its range. Similarly, it is often helpful to restrict a function to a subset of its domain on which the function is injective. Later, we will study the spectral theory of operators. We define the spectrum of an operator T as the set of scalars λ such that $\lambda I - T$ is not invertible. Surely, we will want to know when an operator is invertible—we will also find reason to demonstrate that an inverse is bounded.

Definition 4 (open mapping). *For X, Y metric spaces and $T : X \rightarrow Y$, T is called an open mapping if for any open set $A \subset X$, the image $T(A)$ is an open subset of Y .*

Theorem 8 (open mapping theorem). *If X, Y are Banach spaces and T is a bounded linear map from X onto Y , then T is an open mapping.*

Proof. □

The essential intuitive takeaway from the open mapping theorem is the following:

Corollary 1. *If X is a Banach space and $T : X \rightarrow X$ is a bounded and invertible linear map, then T^{-1} is bounded.*

As a first application, we show that compact operators on infinite-dimensional spaces are not invertible. Remember, this question is only meaningful for complete spaces because compactness requires completeness.

Corollary 2. *If X is a Banach space of infinite dimension and $T : X \rightarrow X$ is a compact linear operator, then T is not invertible.*

Proof. Because T is compact, T is bounded. From here we consider the two possible cases: $\text{range}(T)$ is not closed, and $\text{range}(T)$ is closed.

- If $\text{range}(T)$ is not closed, then $\text{range}(T)$ is not X , since any metric space is a closed subset of itself. So T is not surjective and thus not invertible.
- If $\text{range}(T)$ is closed, then $\text{range}(T)$ is complete as a closed subspace of a Banach space. From here, we can say that by the open mapping theorem, T is an open mapping onto its range. Then the image $T(B_X(0, 1))$ of the open unit ball is an open subset of $\text{range}(T)$. This image contains zero since $T0 = 0$, and because the image is open, it contains an open ball centered at zero:

$$\exists \alpha \in \mathbb{R}, \alpha B_{\text{range}(T)}(0, 1) \subset T(B_X(0, 1)) .$$

Because T is a compact operator, $B_{\text{range}(T)}(0, 1)$ is precompact. From here we conclude by Riesz's theorem that $\text{range}(T)$ has finite dimension. Because X has infinite dimension, T is not surjective and thus not invertible. □

A quicker proof goes as follows. If $T : X \rightarrow X$ is compact and invertible, then T^{-1} is bounded, according to the open mapping theorem. As a composition of a bounded operator with a compact operator, the identity $I = T^{-1}T$ is a compact operator. The image of the unit ball through the identity, namely the unit ball, is thus a precompact set, so by Riesz's theorem, X has finite dimension.

In the approach given first, we give X infinite dimension by assumption and split into two cases. The benefit of running this way is that we see: any compact operator on a Banach space will either have finite rank or it will not have closed range.

Next we consider the closed graph theorem. As a biconditional, it gives us a brand new characterization of continuous linear operators.

Theorem 9 (closed graph theorem). *For X, Y Banach spaces, a linear map $T : X \rightarrow Y$ is continuous if and only if its graph is a closed subspace of $X \times Y$.*

Proof.

- Suppose that T is continuous. Let $\{(x_n, Tx_n)\}_{n \in \mathbb{N}}$ be a convergent sequence in the graph of T , with

$$(x_n, Tx_n) \rightarrow (x, y) \in X \times Y.$$

We want to know if $Tx = y$, telling us that the graph is closed. First, because plucking the first component of the ordered pair is a continuous operation, we have

$$x_n \rightarrow x \in X.$$

By continuity of T , we then have

$$Tx = T\left(\lim_{n \rightarrow \infty} x_n\right) = \lim_{n \rightarrow \infty} (Tx_n) = y.$$

So (x, y) belongs to the graph of T , and thus the graph of T is closed in $X \times Y$.

- Suppose that T has closed graph, and define the continuous, linear projection functions

$$p_1 : (x, y) \mapsto x, \quad p_2 : (x, y) \mapsto y.$$

Define also

$$\pi : \text{graph}(T) \rightarrow X, \pi : (x, Tx) \mapsto x,$$

so that π is the restriction of p_1 to the graph of T . We will show that π has continuous inverse. First, $\text{graph}(T)$ is a Banach space as a closed subspace of the Banach space $X \times Y$. π is injective because $\pi(x, Tx) = 0$ iff $(x, Tx) = (0, 0)$. π is surjective because for any $x \in X$, $\pi(x, Tx) = x$. So π is a bijective, continuous linear map between Banach spaces. By the open mapping theorem, π^{-1} is continuous.

We obtain our conclusion by writing T as a composition of continuous functions. For any $x \in X$,

$$Tx = p_2((x, Tx)) = p_2(\pi^{-1}(x)).$$

□

Notice that the proof does not mention that the spaces X and Y have norms, hinting that the result might have analogues in other settings. However, the proof here does use the open mapping theorem, which we prove using the Baire category theorem. Because Baire's theorem is a metric space result, we can't immediately extend the closed graph theorem to arbitrary topological vector spaces.

The following corollary helps us apply the closed graph theorem to specific operators.

Corollary 3. *For $T : X \rightarrow Y$ a linear map, with X, Y Banach spaces, T is continuous iff $y = 0$ whenever $\{x_n\}_{n \in \mathbb{N}} \subset X$ with $\lim_{n \rightarrow \infty} x_n = 0$ and $\lim_{n \rightarrow \infty} Tx_n = y$.*

For example, we can prove the Hellinger-Toeplitz theorem, which says:

Theorem 10. *For $\mathcal{H}$ a Hilbert space, if $T : \mathcal{H} \rightarrow \mathcal{H}$ is a self-adjoint linear operator, then T is bounded.*

This theorem requires a few new concepts.

- A Hilbert space is a complete normed space, a.k.a. a Banach space, whose norm is induced by an inner product: $\|x\|_{\mathcal{H}} = \sqrt{|\langle x, x \rangle|}$.

- An operator $T : \mathcal{H} \rightarrow \mathcal{H}$ is *self-adjoint* if it can move across the inner product: for any $x, y \in \mathcal{H}$, $\langle Tx, y \rangle = \langle x, Ty \rangle$.
- We need to know that $0 \in \mathcal{H}$ is the *only* element that is orthogonal to everybody else: $\forall x \in \mathcal{H}, \langle 0, x \rangle = 0$.
- Finally, if $x_n \rightarrow x \in \mathcal{H}$, then we also know x_n converges *weakly* to x , denoted $x_n \rightharpoonup x$, which means that for all $z \in \mathcal{H}$, $\langle x_n, z \rangle \rightarrow \langle x, z \rangle$. At least, this definition describes weak convergence in Hilbert spaces. Later, we'll see an equivalent definition for any Banach space.

Now we're ready to prove the theorem.

Proof. Let $\{x_n\}_{n \in \mathbb{N}} \subset \mathcal{H}$ with $\lim_{n \rightarrow \infty} x_n = 0$, and suppose that the limit of Tx_n exists:

$$\lim_{n \rightarrow \infty} Tx_n =: y \in \mathcal{H}.$$

Because $x_n \rightarrow 0$ strongly, $x_n \rightharpoonup x$ weakly as well. So for all $z \in \mathcal{H}$,

$$\lim_{n \rightarrow \infty} \langle x_n, z \rangle = \langle 0, z \rangle = 0.$$

Using continuity of the inner product, and because T is self-adjoint,

$$\begin{aligned} \forall z \in \mathcal{H}, 0 &= \lim_{n \rightarrow \infty} \langle x_n, Tz \rangle \\ &= \lim_{n \rightarrow \infty} \langle Tx_n, z \rangle \\ &= \langle \lim_{n \rightarrow \infty} Tx_n, z \rangle \\ &= \langle y, z \rangle. \end{aligned}$$

From here we observe that for all $z \in \mathcal{H}$, $\langle y, z \rangle = 0$. So $y = 0$, and by corollary 3 obtained from the closed graph theorem, T is bounded. $\square$

Lastly, we apply the open mapping theorem to prove:

Theorem 11 (coercivity equivalence). *For X, Y Banach spaces and $T : X \rightarrow Y$ a bounded linear map, the following statements are equivalent:*

- $\exists c > 0, \forall x \in X, \|Tx\| \geq c\|x\|$ (coercivity),
- T is injective with closed range.

Proof.

- Suppose that the first condition holds:

$$\exists c > 0, \forall x \in X, \|Tx\| \geq c\|x\|.$$

We will not use the open mapping theorem until we prove the other direction. To see that T is injective, let $x \in X$, $x \neq 0$. Then $\|Tx\| \geq c\|x\| > 0$. So $Tx \neq 0$. Thus T is injective.

Now we show that T has closed range. Let $\{x_n\}_{n \in \mathbb{N}} \subset X$ be a sequence, with

$$\lim_{n \rightarrow \infty} Tx_n = y \in Y.$$

Because Tx_n converges, Tx_n is a bounded sequence. Because $\|Tx_n\| \geq c\|x_n\|$, x_n must be bounded as well. By continuity of T , if x_n converges to some x in X , then $y = Tx$, as needed. We will argue that x_n does indeed converge. It helps to draw a picture for this remaining part.

Suppose for contradiction that x_n does not converge. Because x_n does not converge, there is some $\epsilon > 0$ so that each term in the sequence is separated from infinitely many subsequent terms by a distance of at least $3c\epsilon$. Let x_N be a term such that $\|Tx_N - y\| \leq \epsilon$. Let $M > N$, such that $\|x_N - x_M\| > 3c\epsilon$. By the coercivity condition, $\|Tx_N - Tx_M\| \geq 3\epsilon$. By the triangle inequality, $\|Tx_M - y\| > \epsilon$. So Tx_n does not converge to y .

So T must have closed range.

- Now suppose that T is injective with closed range. Because X and Y are Banach spaces and $\text{range}(T)$ is closed, $T : X \rightarrow \text{range}(T)$ is a bijective map between Banach spaces. From the open mapping theorem, $T^{-1} : \text{range}(T) \rightarrow X$ is bounded. Let c^{-1} be the norm of T^{-1} . Let $x \in X$, with $Tx = y$. Then

$$\|x\| = \|T^{-1}y\| \leq c^{-1}\|y\| = c^{-1}\|Tx\|.$$

Rearranging, we see

$$\|Tx\| \geq c\|x\|.$$

□

3.5 Hahn-Banach

Earlier, we mentioned that the open mapping theorem is a nice tool to have on hand when studying the spectral theory of operators. Likewise, when we study the weak and weak-* topologies of normed spaces, we frequently get ourselves out of trouble using the Hahn-Banach theorem. Instead of being defined by a norm like we've seen so far, weak topologies use continuous functionals to declare which sets are open. When we need to find a functional to accomplish a specific goal, the Hahn-Banach theorem often helps us prove the existence of such a functional.

We will need to gather some materials before we can craft the proof. Let us first state the main theorem to see what we're getting into. Later on, we will discuss the theorem's consequences.

Theorem 12 (Hahn-Banach). *Let X be a normed vector space and $F \subset X$ a subspace. Let $\varphi : F \rightarrow \mathbb{R}$ be a bounded linear functional, i.e.*

$$\forall x \in F, |\varphi(x)| \leq \|\varphi\| \cdot \|x\|.$$

Then there exists a continuous linear functional $\Phi : X \rightarrow \mathbb{R}$ defined on the whole space ($\Phi \in X'$), such that

$$\Phi|_F = \varphi \quad \text{and} \quad \|\Phi\| = \|\varphi\|.$$

The theorem says that, given any bounded linear functional φ defined on a subspace of X , φ can be extended to the whole space while maintaining its norm.

There is a more general form of Hahn-Banach that we will use to extract some corollaries. To state it, we will first define an object used in the theorem. In the basic formulation of Hahn-Banach, we assume that φ is continuous, i.e. that

$$\forall x \in F, |\varphi(x)| \leq \|\varphi\| \cdot \|x\|.$$

Now we will relax this requirement to say that φ is bounded by the value of a function p , called a *gauge*.

Definition 5. *A function $p : F \rightarrow \mathbb{R}$ is called a gauge if p is subadditive*

$$\forall x, y \in F, p(x + y) \leq p(x) + p(y),$$

and p is positive homogeneous:

$$\forall \lambda \in \mathbb{R}, \forall x \in F, p(\lambda x) = |\lambda|p(x).$$

Any norm is a gauge. The proof of the following generalized form of Hahn-Banach is essentially the same as the basic form.

Theorem 13 (generalized Hahn-Banach). *Let X be a normed space with $F \subset X$ a subspace. Let $\varphi : F \rightarrow \mathbb{R}$ be a linear function, not necessarily continuous, such that for some gauge p ,*

$$\forall x \in F, \varphi(x) \leq p(x).$$

Then there exists a linear map $\Phi : X \rightarrow \mathbb{R}$ such that

$$\Phi|_F = \varphi \quad \text{and} \quad \forall x \in X, \Phi(x) \leq p(x).$$

Infinite dimension makes problems difficult to visualize. When we discuss Hilbert spaces, we will find that projections and orthonormal bases are helpful in this way. While these tools are unavailable in arbitrary normed spaces, corollaries of the Hahn-Banach theorem give us options to think about the geometry of Banach space, particularly by using linear functionals like the lines on a football field. As a first example, we use Hahn-Banach to prove the following:

Corollary 4. *Let $(X, \|\cdot\|)$ be a Banach space, and let $x_0 \in X$ with $x \neq 0$. Then there exists a continuous linear functional $\varphi \in X'$, such that*

$$\begin{cases} |\varphi(x)| & \leq \|x\| \\ \|\varphi\| & = 1 \\ \varphi(x_0) & = \|x_0\|. \end{cases}$$

Proof. Note that any functional φ satisfying the second criterion will satisfy the first, so we concentrate on the second and third.

As our subspace F , we use the span of x_0 . Define a preliminary functional $\hat{\varphi}$ restricted to F :

$$\hat{\varphi} : x_0 \rightarrow \mathbb{R} \quad \hat{\varphi}(x) = \varphi(ax_0) = a\|x_0\|.$$

We can see that $\hat{\varphi}$ is linear and continuous, and here we check that $\hat{\varphi}$ meets our criteria:

$$\|\hat{\varphi}\| = \sup_{x \in F, \|x\|=1} |\varphi(x)| = \max_{x = -\|x_0\|^{-1}x_0, x = \|x_0\|^{-1}x_0} |\varphi(x)| = 1,$$

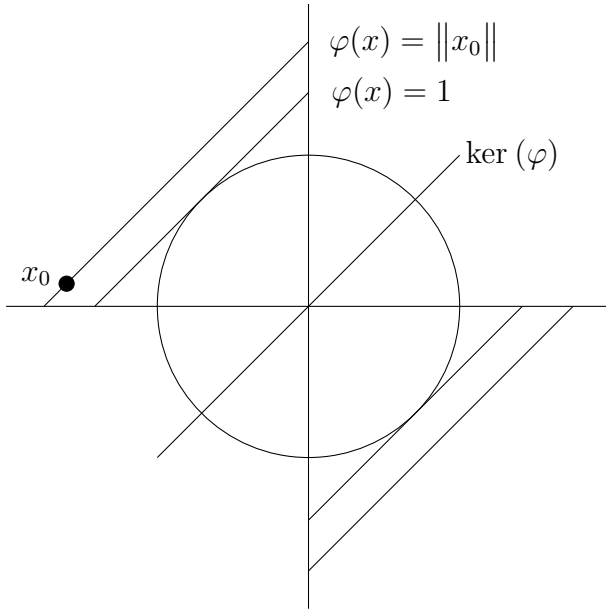


Figure 1: Linear functionals draw parallel isoclines.

and

$$\hat{\varphi}(x_0) = \|x_0\|.$$

By Hahn-Banach, there is continuous linear functional $\varphi \in X'$ with

$$\varphi|_F = \hat{\varphi} \quad \text{and} \quad \|\varphi\| = \|\hat{\varphi}\|.$$

Thus

$$\|\varphi\| = 1 \quad \text{and} \quad \varphi(x_0) = \|x_0\|.$$

□

A geometric takeaway is that linear functionals draw parallel isoclines, and we will continue to justify it with further results. This visual will be critical in the consideration of weak topologies.

Giving us more geometry, the next corollary says that you can wedge a linear functional between any distinct points of a linear space.

Corollary 5. *Let X be a normed linear space. Then X' separates the space: for any $x_1, x_2 \in X$, $x_1 \neq x_2$,*

$$\exists \varphi \in X', \varphi(x_1) < 0, \varphi(x_2) > 0.$$

Proof. In the spirit of Hahn-Banach, we build a linear functional $\hat{\varphi}$ on a subspace:

$$\begin{aligned} \hat{\varphi} : \text{span}\{x_1, x_2\} &\rightarrow \mathbb{R} \\ ax_1 + bx_2 &\mapsto b - a. \end{aligned}$$

We see that

$$\begin{aligned} \hat{\varphi}(x_1) &= -1 < 0 \\ \hat{\varphi}(x_2) &= 1 > 0. \end{aligned}$$

As a linear map on a finite-dimensional space, $\hat{\varphi}$ is bounded. Hahn-Banach tells us that there is an extension

$$\varphi \in X', \quad \varphi|_{\text{span}\{x_1, x_2\}} = \hat{\varphi}.$$

□

Alternatively, we could have proved this result using the previous corollary:

$$\exists \varphi \in X', \quad \varphi(x_2 - x_1) = \|x_2 - x_1\|.$$

Corollary 6. *Let X be a normed linear space. Then*

$$\forall x \in X, \quad \|x\| = \sup_{\|\varphi\| \leq 1} |\varphi(x)|.$$

Proof. From the definition of norm in the space X' no functional with norm 1 can take absolute value greater than $\|x\|$ when evaluated on x . However, the norm is achieved. From corollary 4, we know that there is a functional $\varphi_x \in X'$ with $\|\varphi_x\| = 1$ and $\varphi(x) = \|x\|$. □

The next result has more of an algebraic than geometric character. The dual of the dual of a space is called the bidual. When a space is isomorphic to its bidual, it is called *reflexive*. Not all spaces X are reflexive, but as we see here, X is always at least isomorphic to a particular subspace of X'' :

Theorem 14. *The following function, called the canonical injection, is an isometry:*

$$\begin{aligned}\delta : X &\rightarrow X'' \\ x &\mapsto \delta_x,\end{aligned}$$

where for any $\varphi \in X'$, $\delta_x(\varphi) = \varphi(x)$.

Proof. Let $x \in X$. We need to check that

$$\|x\|_X = \|\delta_x\|_{X''}.$$

From corollary 6, we know that

$$\|x\|_X = \sup_{\|\varphi\|=1} |\varphi(x)|.$$

This result plugs into the definition of norm in X'' :

$$\begin{aligned}\|\delta_x\|_{X''} &:= \sup_{\|\varphi\|=1} |\delta_x(\varphi)| \\ &= \sup_{\|\varphi\|=1} |\varphi(x)| \\ &= \|x\|.\end{aligned}$$

□

When we ask if a space is reflexive, we're trying to see if there is a functional in X'' whose action amounts to more than just evaluation at a point in X .

The next theorem is a bit of a linear space version of Urysohn's lemma from topology.

Theorem 15. *Let X be a normed linear space, $F \subset X$ a closed subspace, and $x_0 \notin F$. Then $\exists \Phi \in X'$ such that*

$$\Phi|_F = 0, \Phi(x_0) = 1 \text{ and } \|\Phi\| = \delta^{-1},$$

where $\delta = \text{dist}(x_0, F)$.

Proof. Let $V = \text{span}\{x_0\} \oplus F$, and define

$$\begin{aligned}\varphi : V &\rightarrow \mathbb{R} \\ ax_0 + f &\mapsto a,\end{aligned}$$

with a a scalar and $f \in F$. Then we can show that φ is continuous and linear, and we see right away that

$$\varphi(x_0) = 1 \text{ and for all } f \in F, \varphi(f) = 0.$$

It remains to track down that $\|\varphi\| = \delta^{-1}$.

- To show that $\|\varphi\| \geq \delta^{-1}$, we consider the action of φ on x_0 . Let $f \in F$. Then

$$\begin{aligned}\|\varphi(x_0)\| &= 1 \\ \|\varphi(x_0) - \varphi(f_2)\| &= 1 \\ \|x_0 - f_2\| &\geq \delta,\end{aligned}$$

so

$$\varphi \geq \frac{\|\varphi(x_0) - \varphi(f)\|}{\|x_0 - f\|} = \delta^{-1}.$$

- Now to show that $\|\varphi\| \leq \delta^{-1}$. Let $x = ax_0 + f$, $f \in F$. Then

$$d(ax_0, F) = \inf_{f \in F} \|ax_0 - f\| = |a| \cdot \inf_{w \in F} \|x_0 - w\|.$$

We can also observe that

$$|\varphi(x)| = |a| = \frac{d(x, F)}{d(x_0, F)}.$$

So we can bound $\|\varphi\|$ by substituting:

$$\begin{aligned}
\|\varphi\| &= \sup_{x \in V} \frac{\|\varphi(x)\|}{\|x\|} \\
&= \frac{|\varphi(ax_0 + f)|}{\|ax_0 + f\|} \\
&\leq \frac{|a| \cdot \|\varphi(x_0 + f)\|}{\|ax_0 + f\|} \\
&= \frac{|a|}{\|ax_0 + f\|} \\
&\leq \frac{|a|}{|a| \cdot \delta} \\
&= \delta^{-1}.
\end{aligned}$$

We didn't verify linearity, but if we did, this result would give us continuity as well. Now that $\|\varphi\| = \delta^{-1}$, it can be extended by Hahn-Banach to $\Phi \in X'$, with $\Phi|_V = \varphi$ and $\|\Phi\| = \|\varphi\|$.

□

3.6 Banach-Steinhaus

As another application of Baire's theorem, the uniform boundedness principle and its extension as the Banach-Steinhaus theorem demonstrate the utility of functional analysis in concrete settings. The uniform boundedness principle says that if a collection of bounded linear maps from a Banach space to a normed space is pointwise bounded, it is uniformly bounded.

Theorem 16 (uniform boundedness principle). *For X a Banach space, Y a normed space, and $T_i \in \mathcal{B}(X, Y)$, $i \in I$ a collection of bounded linear maps, if*

$$\forall x \in X, \exists c_x \geq 0, \|T_n x\| \leq c_x \forall n \in I,$$

then

$$\exists c > 0, \|T_i\| \leq c \forall i \in I.$$

Proof. We will use the pointwise boundedness of $T_{ii \in I}$ to define a collection of subsets that covers all of X . Because X is complete, we can use Baire's theorem to extract a critical realization. The remainder of the proof will employ a scaling argument to bound $\{\|T_i\|\}_{i \in I}$.

The pointwise boundedness of $\{T_i\}_{i \in I}$ allows us to define for all $n \in \mathbb{N}$

$$\Phi_n = \{x \in X : \forall i \in I, \|T_i x\| \leq n\}$$

and to say that X is their union:

$$X = \bigcup_{n=0}^{\infty} \Phi_n.$$

To apply Baire's theorem, we need to check that each Φ_n is closed. Let us rephrase the definition of Φ_n :

$$\Phi_n = \bigcap_{i \in I} T_i^{-1}(B(0, n)).$$

If we take $B(0, n)$ to be the *closed* unit ball, then because T_i is continuous, it pulls closed sets back to closed sets. Because arbitrary intersections of closed sets are closed, Φ_n is closed.

Baire's theorem says that, in a complete space, a countable union of closed sets with empty interior has empty interior. Because X is this union, and X does not have empty interior, at least one of the Φ_N in the union has nonempty interior. At this point, we conclude the proof by employing a scaling argument.

Let $x_0 \in \text{int}(\Phi_N)$. Then

$$\exists \epsilon > 0, \forall x \in B(x_0; \epsilon), \forall i \in I, \|T_i x\| \leq N.$$

Now let $x \in B(0, \epsilon)$. By triangle inequality,

$$\begin{aligned} \forall i \in I, \|T_i x\| &= \|T_i(x_0 + x) - T_i(x_0)\| \\ &\leq \|T_i(x_0 + x)\| + \|T_i(x_0)\| \\ &\leq N + N \\ &= 2N. \end{aligned}$$

Since $\|x\| \leq \epsilon$,

$$\frac{\|T_i x\|}{\|x\|} \leq \frac{2N}{\epsilon}.$$

Since x is an arbitrary element of $B(0, \epsilon)$, this ratio applies to any $x \in X$. So

$$\forall i \in I, \|T_i\| \leq \frac{2N}{\epsilon}.$$

□

On the surface, this theorem appears to be a way to tell when a collection of maps is uniformly bounded. That is true, and you're welcome to use the theorem this way, for instance to apply the Arzelà-Ascoli theorem. Some of the more satisfying application of the uniform boundedness principle is though an indirect argument. For example, notice that the theorem requires the codomain of the maps to be a Banach space. If you want to show that a given space X is not complete, you might be able to design a sequence of maps from X into *some* normed space Y . If it turns out that the sequence of maps is pointwise bounded but not uniformly bounded, then X cannot possibly be complete.

The following example gives another way to reverse conclusions out of the uniform boundedness principle, in this case to prove the existence of some object.

Example 1. *The Fourier series is a way to represent a function on a compact interval, such as $[0, 2\pi]$, as an infinite series, and it tends to work even when the function is not continuous. Surprisingly, there are continuous functions $f \in \mathcal{C}([0, 2\pi])$ whose Fourier series do not converge to $f(x)$ at every point. With some work, you can show that the partial sum $T_n f$ of the Fourier coefficients is not a uniformly bounded operation. That means T_n is not pointwise bounded either, so there must be some function $f \in \mathcal{C}([0, 2\pi])$ whose Fourier series does not converge at $x = 0$.*

We are continuing a theme here. Hahn-Banach gives us ways to inspect a normed space using linear functionals as tools, and now, so does the uniform boundedness principle.

The following two results are both interchangeably referred to as *Banach-Steinhaus*, where the second is a stronger version of the first. They both apply the uniform boundedness principle to say what happens when the maps are pointwise *convergent*, not just bounded. Because they describe a linear map obtained in the limit, they require the codomain to be a Banach space, not just the domain.

Corollary 7 (Banach-Steinhaus). *Let X, Y be Banach spaces, with $\{T_n\}_{n \in \mathbb{N}} \subset \mathcal{B}(X, Y)$ a sequence of bounded linear maps such that*

$$\forall x \in X, \lim_{n \rightarrow \infty} T_n x = T x.$$

Then

1. T is linear;
2. $T \in \mathcal{B}(X, Y)$;
3. $\sup_{n=0}^{\infty} \|T_n\| < \infty$;
4. $\|T\| \leq \liminf_{n \rightarrow \infty} \|T_n\|$.

The corollary above doesn't discuss the rate at which convergence occurs. The next result gives us a little more:

Theorem 17 (Strong Form of Banach-Steinhaus). *For X, Y Banach spaces and $\{T_n\}_{n \in \mathbb{N}} \subset \mathcal{B}(X, Y)$ a sequence of bounded linear maps such that*

$$\forall x \in X, \lim_{n \rightarrow \infty} T_n x = T x,$$

then $\{T_n\}_{n \in \mathbb{N}}$ converges uniformly to T on every compact set $K \subset X$, and $T \in \mathcal{B}(X, Y)$.

We will use Banach-Steinhaus to prove that weakly convergent sequences are bounded.

4 Weak Topologies

4.1 Weak Convergence

So far we have discussed spaces in which the topology arises from a metric, and more specifically, a norm. Sometimes this topology is too restrictive, such that certain problems in optimization and mathematical physics have no solution. In this chapter, we explore a number of ways to relax the topology on our space, such that we can reframe our problems to have solutions.

In our study of metric spaces, we explore the connections between topology and the convergence of sequences. In this first section, we introduce a new notion of convergence.

Definition 6. For X a normed space, a sequence $\{x_n\}_{n \in \mathbb{N}} \subset X$ converges weakly to $x \in X$, denoted $x_n \rightharpoonup x$, if for all $\varphi \in X'$,

$$\lim_{n \rightarrow \infty} \varphi(x_n) = \varphi(x).$$

First we check for an important property. Thankfully, any weakly convergent sequence is bounded:

Theorem 18. If X is a normed space and $x_n \rightharpoonup x \in X$, then $\{x_n\}_{n \in \mathbb{N}}$ is bounded, and $\|x\| \leq \liminf_{n \rightarrow \infty} \|x_n\|$.

Proof. Let $\{x_n\}_{n \in \mathbb{N}} \subset X$ be a sequence, with $x_n \rightharpoonup x \in X$. Then

$$\forall \varphi \in X', \lim_{n \rightarrow \infty} \varphi(x_n) = \varphi(x).$$

Define for all $x \in X$ the composition operator

$$\begin{aligned} \delta_x : X' &\rightarrow \mathbb{R}, \\ \varphi &\mapsto \varphi(x), \end{aligned}$$

and consider the sequence $\{\delta_{x_n}\}_{n \in \mathbb{N}}$. For any $\varphi \in X'$, because $x_n \rightharpoonup x$,

$$\lim_{n \rightarrow \infty} \delta_{x_n}(\varphi) = \lim_{n \rightarrow \infty} \varphi(x_n) = \varphi(x) = \delta_x(\varphi).$$

So $\{\delta_{x_n}\}$ converges pointwise to δ_x over X' . Because X' and $\mathbb{R}$ are complete, by Banach-Steinhaus,

$$\delta_x \in X'', \quad M := \sup_{n \in \mathbb{N}} \|\delta_{x_n}\| \leq \infty, \quad \text{and} \quad \|\delta_x\| \leq \liminf_{n \in \mathbb{N}} \|\delta_{x_n}\|.$$

This result is useful because we can show that $\|\delta_{x_n}\| = \|x_n\|$. First,

$$\|\delta_{x_n}\| = \sup_{\|\varphi\|=1} |\delta_{x_n}(\varphi)| = \sup_{\|\varphi\|=1} |\varphi(x_n)| \leq \sup_{\|\varphi\|=1} \|\varphi\| \cdot \|x_n\| = \|x_n\|.$$

To show that the norm is attained, we use a corollary of Hahn-Banach to say:

$$\exists \varphi \in X', \quad \varphi(x_n) = \|x_n\|, \quad \|\varphi\| = 1,$$

and so

$$\|\delta_{x_n}\| \geq |\varphi(x_n)| = \|x_n\|.$$

Thus $\|\delta_{x_n}\| = \|x_n\|$.

From the result found via Banach-Steinhaus, $\|\delta_{x_n}\|$ is bounded, so $\|x_n\|$ is bounded. Also

$$\left(\|\delta_x\| \leq \liminf_{n \in \mathbb{N}} \|\delta_{x_n}\| \right) \text{ implies } \left(\|x\| \leq \liminf_{n \in \mathbb{N}} \|x_n\| \right).$$

□

4.2 Why Weak Topology

5 Hilbert Space

5.1 Basic Definitions

Here we define the inner product:

Definition 7. *If E is a vector space with $B : E \times E \rightarrow \mathbb{C}$ positive sesquilinear, then E together with B is called a pre-Hilbert space, or alternatively, an inner product space.*

Lemma 3. *For E a vector space and B a nonnegative sesquilinear form, the map*

$$\begin{aligned} \|\cdot\| : E &\rightarrow [0, \infty) \\ x &\mapsto \sqrt{B(x, x)} \end{aligned}$$

is a semi-norm. If B is positive definite, then the map is a norm.

Typically, we will not use B to refer to the inner product but instead use $\langle \cdot, \cdot \rangle$.

Example 2. *For $\mathbb{R}^n$ and $A \in \mathbb{R}^{n \times n}$ nonsingular, the map*

$$x \mapsto (Ax)^T(Ax) = x^T(A^T A)x$$

is an inner product.

Example 3.

$$\ell^2 = \left\{ (x_n)_{n \in \mathbb{N}} : x_n \in \mathbb{C}, \sum_0^\infty |x_n|^2 < \infty \right\}$$

for $x, y \in \ell^2$, the map

$$\langle x, y \rangle = \sum_0^\infty x_n \overline{y_n}$$

is an inner product.

Theorem 19. *Every vector space has an inner product.*

Proof. Let $\{e_i\}_{i \in I}$ be an algebraic (Hamel) basis of E . For any $x, y \in E$ we have the representations

$$\begin{aligned} x &= \sum_{i \in I_0} x_i e_i, \quad |I_0| < \infty \\ y &= \sum_{j \in J_0} y_j e_j, \quad |J_0| < \infty, \end{aligned}$$

with $J := I_0 \cup J_0$. Then we obtain an inner product from

$$\langle x, y \rangle = \sum_{i \in J} x_i y_i.$$

□

We will basically never use this last result, because Hamel bases are usually impossible to specify concretely, and their cardinality gets out of hand really quickly.

Theorem 20 (Cauchy-Schwarz). *Given $(E, \langle \cdot, \cdot \rangle)$ a pre-Hilbert space, for all $x, y \in E$,*

$$|\langle x, y \rangle|^2 \leq \langle x, x \rangle \cdot \langle y, y \rangle.$$

Proof. The proof relies on a somewhat arbitrary starting point and a difficult substitution partway through. It all hinges on the nonnegativity of the inner product: an inner product of a vector with itself is always nonnegative. For all $\lambda, \mu \in \mathbb{C}$,

$$\begin{aligned} 0 &\leq \langle \lambda x - \mu y, \lambda x - \mu y \rangle \\ &= |\lambda|^2 \langle x, x \rangle - \langle \lambda x, \mu y \rangle - \langle \mu y, \lambda x \rangle + |\mu|^2 \langle y, y \rangle \\ \langle \lambda x, \mu y \rangle + \overline{\langle \lambda x, \mu y \rangle} &\leq |\lambda|^2 \langle x, x \rangle + |\mu|^2 \langle y, y \rangle \\ \lambda \bar{\mu} \langle x, y \rangle + \bar{\lambda} \mu \overline{\langle x, y \rangle} &\leq |\lambda|^2 \langle x, x \rangle + |\mu|^2 \langle y, y \rangle. \end{aligned}$$

Now we substitute

$$\lambda = \|y\| \cdot \langle x, y \rangle \cdot |\langle x, y \rangle|^{-1}, \quad \mu = \|x\|$$

and simplify:

$$\begin{aligned}
& \|y\| \langle x, y \rangle |\langle x, y \rangle|^{-1} \cdot \|x\| \langle x, y \rangle + \|y\| \overline{\langle x, y \rangle} |\langle x, y \rangle|^{-1} \cdot \|x\| \overline{\langle x, y \rangle} \\
& \leq \langle y, y \rangle |\langle x, y \rangle|^2 (|\langle x, y \rangle|^{-1})^2 \langle x, x \rangle + \langle x, x \rangle \langle y, y \rangle \\
& 2\|y\| \cdot \|x\| |\langle x, y \rangle| \leq 2\langle y, y \rangle \langle x, x \rangle \\
& |\langle x, y \rangle| \leq \|y\|^{-1} \|x\| \langle x, x \rangle \langle y, y \rangle \\
& |\langle x, y \rangle|^2 \leq \frac{|\langle x, x \rangle|^2 |\langle y, y \rangle|^2}{|\langle x, x \rangle| \cdot |\langle y, y \rangle|} \\
& |\langle x, y \rangle|^2 \leq \langle x, x \rangle \langle y, y \rangle.
\end{aligned}$$

□

Lemma 4. For $(E, \langle \cdot, \cdot \rangle)$ a pre-Hilbert space, the following functions are continuous:

$$\begin{aligned}
x_0 : E &\rightarrow \mathbb{C} \\
y &\mapsto \langle x_0, y \rangle
\end{aligned}$$

and

$$\begin{aligned}
y_0 : E &\rightarrow \mathbb{C} \\
x &\mapsto \langle x, y_0 \rangle.
\end{aligned}$$

Proof. Use Cauchy-Schwarz. □

Definition 8. We call a pre-Hilbert space $(E, \langle \cdot, \cdot \rangle)$ a Hilbert space if E is complete for the topology induced by

$$\|x\| = \sqrt{\langle x, x \rangle}. \quad (1)$$

Example 4. The vector space $\mathcal{C}[0, 1]$ of complex-valued continuous functions on $[0, 1]$ with the inner product

$$\langle f, g \rangle = \int_0^1 f(t) \overline{g(t)} dt$$

is pre-Hilbert but not Hilbert, because it is not complete in the induced norm.

Definition 9. We call elements x and y of a pre-Hilbert space orthogonal if

$$\langle x, y \rangle = 0.$$

Lemma 5. For $\{x_i\}_{i=1,\dots,N}$ a finite family of elements in E that are pairwise orthogonal ($\langle x_i, x_j \rangle = \delta_{ij}$), then

$$\left\| \sum_{i=1}^N x_i \right\|^2 = \sum_{i=1}^N \|x_i\|^2.$$

Not all norms are derived from an inner product. For example, the normed space $(\mathcal{C}[0,1], \|\cdot\|_\infty)$ does not arise from an inner product. However, the following result and proof from [?] tells us when the equivalence holds.

Theorem 21 (parallelogram law). A normed linear space X is an inner product space with a norm derived from the inner product as in Eq. 1 if and only if

$$\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2 \quad \text{for all } x, y \in X.$$

Definition 10. For W a subspace of a Hilbert space E , we define the orthogonal complement of W by

$$W^\perp := \{x \in E : \forall w \in W, \langle x, w \rangle = 0\}.$$

Theorem 22. For $\mathcal{H}$ a Hilbert space and $A \subset \mathcal{H}$ a subset, $A^\perp$ is a closed linear subspace of $\mathcal{H}$.

Proof. To show that $A^\perp$ is closed, use the continuity of the inner product. Let $\{x_n\} \subset A^\perp$, with $x_n \rightarrow x \in \mathcal{H}$ as $n \rightarrow \infty$, and let $a \in A$. Then

$$0 = \lim_{n \rightarrow \infty} \langle x_n, a \rangle = \langle \lim_{n \rightarrow \infty} x_n, a \rangle = \langle x, a \rangle.$$

So $A^\perp$ is closed. To show that $A^\perp$ is a linear subspace, let $x, y \in A^\perp$, $a \in A$, and $c, d \in \mathbb{C}$. Then

$$\langle cx + dy, a \rangle = c\langle x, a \rangle + d\langle y, a \rangle = 0.$$

So $cx + dy \in A^\perp$. Note also $0 \in A^\perp$. □

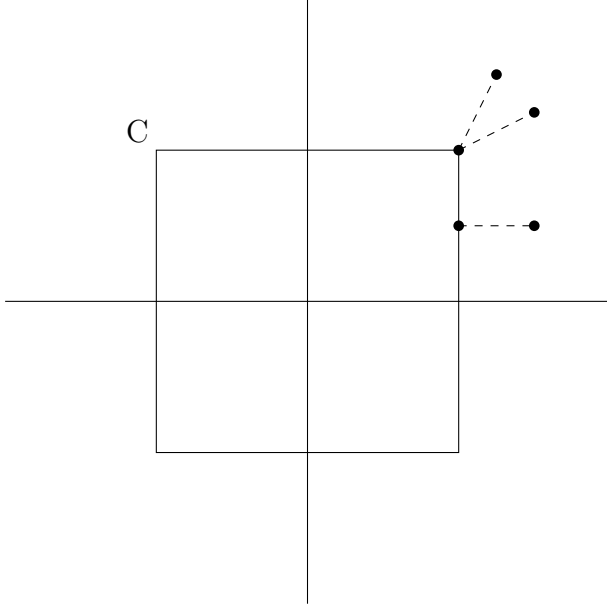


Figure 2: Projections onto convex sets are not necessarily linear or injective.

5.2 Orthogonal Projection

Theorem 23. *Let H be a Hilbert space, $C \subset H$ with C convex and closed in H . Then for all x in H ,*

$$\exists! y \in C, \quad \|x - y\| = \text{dist}(x, C).$$

We call $y = P_C(x)$ the orthogonal projection of x onto C .

Note from Figure 2 that $P_C(x)$ is nonlinear and noninjective.

Proof of projection theorem:

Proof. Let $d := \text{dist}(x, C) = \inf_{v \in C} \|x - v\|$. If $d = 0$, then because C is closed, $x \in C$. For $d > 0$,

$$\forall n \in \mathbb{N}, \exists x_n \in C, \|x - x_n\|^2 \leq d^2 + \frac{1}{n+1}. \quad (2)$$

To show that x_n converges, and that x_n must converge to a unique $y \in C$, let $\{x_p\}_{p \in \mathbb{N}}$ be another sequence with the properties of Eq. 2. Now use the

parallelogram law for $(x - x_n)$ and $(x - x_p)$:

$$\begin{aligned} \|(x - x_n) + (x - x_p)\|^2 + \|(x - x_n) - (x - x_p)\|^2 &= 2 \left(\|x - x_n\|^2 + \|x - x_p\|^2 \right) \\ 4 \left\| x - \frac{x_n + x_p}{2} \right\|^2 + \|x_n - x_p\|^2 &= 2 \left(\|x - x_n\|^2 + \|x - x_p\|^2 \right). \end{aligned}$$

By convexity, $\frac{1}{2}(x_n + x_p) \in C$, so $\|x - \frac{x_n + x_p}{2}\| \geq d$, and thus

$$\|x_n - x_p\|^2 \leq 2 \left(d^2 + \frac{1}{n+1} + d^2 + \frac{1}{p+1} \right) - 4d^2 = 2 \left(\frac{1}{n+1} + \frac{1}{p+1} \right).$$

So we have

$$\lim_{n,p \rightarrow \infty} \|x_n - x_p\| = 0,$$

and if the limit of x_n and x_p exist, they are the same. This argument also tells us that x_n is a Cauchy sequence, and because H is a Hilbert space, x_n converges to some $\tilde{x} \in H$.

Because $d^2 \leq \|x - x_n\| \leq d^2 + \frac{1}{n+1}$, we have

$$d^2 \leq \|x - \tilde{x}\|^2 \leq d^2,$$

so $\|x - \tilde{x}\| = \text{dist}(x, C)$.

□

Lemma 6 (polarization identity, “super” Pythagorean theorem, law of cosines).
For x, y belonging to a pre-Hilbert space,

$$\|x - y\|^2 = \|x\|^2 + \|y\|^2 - 2\Re\langle x, y \rangle.$$

Proof. Express the left side as an inner product and expand:

$$\begin{aligned} \|x - y\|^2 &= \langle x - y, x - y \rangle \\ &= \langle x, x \rangle - \langle x, y \rangle - \langle y, x \rangle + \langle y, y \rangle \\ &= \langle x, x \rangle - \langle x, y \rangle - \overline{\langle x, y \rangle} + \langle y, y \rangle \\ &= \|x\|^2 + \|y\|^2 - 2\Re\langle x, y \rangle. \end{aligned}$$

□

Lemma 7. Let $(X, \|\cdot\|)$ be a normed space. $\|\cdot\|^2$ is a convex function:

$$\forall x, y \in X, \forall t \in [0, 1], \|tx + (1-t)y\|^2 \leq t\|x\|^2 + (1-t)\|y\|^2.$$

By induction, we can say the same for any convex combination in X :

$$\forall \{x_1, \dots, x_k\} \subset X, t_i \geq 0, \left\| \sum_{i=1}^k t_i x_i \right\|^2 \leq \sum_{i=1}^k t_i \|x_i\|^2.$$

Proof. Let $x, y \in X$, and let $t \in [0, 1]$.

$$\begin{aligned} \|tx + (1-t)y\|^2 &= \langle tx + (1-t)y, tx + (1-t)y \rangle \\ &= \langle tx, tx \rangle + \langle tx, (1-t)y \rangle + \langle (1-t)y, tx \rangle + \langle (1-t)y, (1-t)y \rangle \\ &= t^2\|x\|^2 + t(1-t)\langle x, y \rangle + t(1-t)\langle y, x \rangle + (1-t)^2\|y\|^2 \\ &= t^2\|x\|^2 + 2t(1-t)\langle x, y \rangle + (1-t)^2\|y\|^2 \\ &= t^2\|x\|^2 + t(1-t) \left(\|x\|^2 + \|y\|^2 - \|x-y\|^2 \right) + (1-t)^2\|y\|^2 \text{ law of cosines} \\ &\leq \|x\|^2(t^2 + t - t^2) + \|y\|^2(t - t^2 + 1 - 2t + t^2) \\ &= t\|x\|^2 + (1-t)\|y\|^2. \end{aligned}$$

□

Lemma 8. For C a convex subset of a pre-Hilbert space $\mathcal{H}$, for all $x \in \mathcal{H}$ and for all $x_0 \in C$, the following are equivalent:

- (1) x_0 is the projection of x onto C .
- (2) $\Re(\langle x - x_0, y - x_0 \rangle) \leq 0$ for all $y \in C$.
- (3) If C is a subspace of $\mathcal{H}$, then $x - x_0 \in C^\perp$.

Proof. We will show that (1) and (2) are equivalent, and we will show that (2) and (3) are equivalent.

- To show (1) $\Rightarrow$ (2), because $x_0 \in C$ and $y \in C$, we have

$$v = (1-t)x_0 + ty \in C \quad \forall t \in [0, 1].$$

Then by the minimization property of the projection,

$$\begin{aligned}
\|x - x_0\| &\leq \|x - v\| \\
&= \|x - [(1 - t)x_0 + ty]\| \\
&= \|(x - x_0) - t(y - x_0)\| \\
\|x - x_0\|^2 &\leq \langle (x - x_0) - t(y - x_0), (x - x_0) - t(y - x_0) \rangle \\
&= \|x - x_0\|^2 - 2t\Re\langle x - x_0, y - x_0 \rangle + t^2\|y - x_0\|^2 \\
2\Re\langle x - x_0, y - x_0 \rangle &\leq t\|y - x_0\|^2 \quad \forall t \in (0, 1].
\end{aligned}$$

Letting $t \rightarrow 0$, we obtain that $\Re\langle x - x_0, y - x_0 \rangle \leq 0$.

- (3) $\Rightarrow$ (2)

Suppose that $\forall y \in C$, $\langle x - x_0, y \rangle = 0$. Since $x_0 \in C$, with C a vector space,

$$\begin{aligned}
\langle x - x_0, y - x_0 \rangle &= 0 \\
\Rightarrow \Re\langle x - x_0, y - x_0 \rangle &\leq 0.
\end{aligned}$$

- (2) $\Rightarrow$ (3)

We will use the super Pythagorean theorem of Lemma 6, which says that

$$\forall x, y, x_0, \|x - y\|^2 = \|x - x_0\|^2 + \|y - x_0\|^2 - 2\Re(\langle x - x_0, y - x_0 \rangle).$$

Suppose that $\forall y \in C$ and $\forall x_0 \in C$,

$$\Re\langle x - x_0, y - x_0 \rangle \leq 0.$$

Put $z := y - x_0 \in C$. For all such z ,

$$\Re\langle x - x_0, z \rangle \leq 0.$$

Recall that the product of complex conjugates is nonnegative:

$$0 \leq \overline{\langle x - x_0, z \rangle} \langle x - x_0, z \rangle,$$

but also

□

5.3 Riesz Representation

In studying a Banach space X , we consider the dual space of X and use it to say a great many things about X . In Hilbert space, this viewpoint is still valid, and we find that the interpretation simplifies. With a few results, we show that a Hilbert space is isomorphic to its dual. A member of the dual space corresponds to a unique element of the original space, acting on elements through the inner product. An especially nice consequence is that we can consider the adjoint of an operator on $\mathcal{H}$ as another operator on $\mathcal{H}$, rather than a function that acts on a dual space—so much simpler. The Riesz representation theorem is the key player in this development.

Theorem 24 (Riesz representation theorem). *For $\mathcal{H}$ a Hilbert space and $\mathcal{H}'$ its topological dual, for all $\varphi \in \mathcal{H}'$,*

$$\exists! v \in \mathcal{H}, \forall x \in \mathcal{H}, \varphi(x) = \langle x, v \rangle .$$

As a result, $\|\varphi\|_{\mathcal{H}'} = \|v\|_{\mathcal{H}}$.

Proof. In our discussion of Hahn-Banach, we saw that level sets of linear functionals define co-dimension 1 hyperplanes of the space. We will develop and use that idea here.

To start the proof, note that if $\varphi = 0$, then $v = 0$, so assume that $\varphi \neq 0$. As the kernel of a nonzero functional, we know that $\ker(\varphi)$ is a closed, proper subspace of $\mathcal{H}$, and

$$\mathcal{H} = \ker(\varphi) \oplus (\ker(\varphi))^\perp .$$

The remainder of the proof falls into two parts.

- Claim: $W \triangleq (\ker(\varphi))^\perp$ has dimension 1.

Let $x, y \in (\ker(\varphi))^\perp$, $x, y \neq 0$. We can show that x and y are linearly dependent. For $a, b \in \mathbb{C}$,

$$\begin{aligned} ax + by &= 0 \\ \varphi(ax + by) &= 0 \\ a\varphi(x) + b\varphi(y) &= 0 . \end{aligned}$$

This equation is solved by $a = -\varphi(y)$, $b = \varphi(x)$, which are both nonzero because $x, y \in (\ker(\varphi))^\perp$. Thus x and y are linearly dependent, and $\dim W = 1$.

- Now we construct the dual element of φ , which we will call v . Because W has dimension 1,

$$\exists u \in \mathcal{H}, \|u\| = 1, W = \text{span}\{u\}.$$

Let $v = \overline{\varphi(u)}u$. To show that v does the trick, first let $x \in W$. Then

$$\begin{aligned} x &= \alpha u, \alpha = \langle x, u \rangle. & \varphi(x) &= \varphi(\alpha u) = \alpha \varphi(u) = \langle x, u \rangle \varphi(u) \\ \langle x, v \rangle &= \varphi(u) \langle x, u \rangle = \varphi(x), \end{aligned}$$

as needed. For $x \in \ker(\varphi)$, $x \in \ker(\varphi)$, so

$$\langle x, v \rangle = \varphi(u) \langle x, u \rangle = 0 = \varphi(x).$$

Because $\mathcal{H} = \ker(\varphi) \oplus W$, $\varphi(x) = \langle x, v \rangle$ for all $x \in \mathcal{H}$.

To address the issue of uniqueness, note that we could only have chosen u or its negation. If we chose $-u$, our definition of v would give us the very same v .

□

5.4 Orthogonal Bases

Theorem 25. For $\mathcal{H}$ a Hilbert space and V a closed linear subspace, $\mathcal{H} = V \oplus V^\perp$.

Proof. Let $x \in \mathcal{H}$. Because V is a closed linear subspace, we can use the projection theorem. Let $v = \mathcal{P}_V(x)$. Then by the theorem, $v \in V$ and $(x - v) \in V^\perp$. So

$$x = v + (x - v),$$

with $x \in V$ and $x - v \in V^\perp$. So $x \in V \oplus V^\perp$.

Note that the orthogonal direct sum is only defined for a pair of closed linear subspaces. V is a closed linear subspace by assumption, and we know that taking the orthogonal complement of any subset of a Hilbert space yields a closed linear subspace. So we are in good standing.

□

Theorem 26. For $(\mathcal{H}, \langle \cdot, \cdot \rangle)$ a Hilbert space and $\{F_i\}_{i \in I}$ a family of closed, pairwise orthogonal subspaces of $\mathcal{H}$, let $P_i : \mathcal{H} \rightarrow F_i$ be the orthogonal projector onto F_i . Then the following results hold:

- (1) $\forall x \in \mathcal{H}, \sum_{i \in I} \|P_i x\|^2 \leq \|x\|^2$ (Bessel's inequality)
- (2) With $V := \overline{\text{span} \{ \cup_{i \in I} F_i \}}$, the following properties are equivalent:
 - i) $x \in V$
 - ii) $x = \sum_{i \in I} P_{F_i}(x)$
 - iii) $\|x\|^2 = \sum_{i \in I} \|P_i x\|^2$
- (3) $\forall x, y \in V, \langle x, y \rangle = \sum_{i \in I} \langle P_i x, P_i y \rangle$.

6 Fourier Series

7 Spectral Theory in Banach and Hilbert Space

7.1 Hilbert Space Operators

There is some taxonomy of operators on Hilbert space that we will need before jumping into spectral theory.

Definition 11. $T : \mathcal{H} \rightarrow \mathcal{H}$ is self-adjoint if $A = A^*$.

Definition 12. $T : \mathcal{H} \rightarrow \mathcal{H}$ is normal if $TT^* = T^*T$.

Definition 13. $T : \mathcal{H} \rightarrow \mathcal{H}$ is positive if for all $x \in \mathcal{H}, \langle x, Tx \rangle \geq 0$.

Note that the above definition applies to complex Hilbert spaces, not just real ones.

There is a connection between positivity and self-adjointness when the space is complex. This result does not always hold when $\mathcal{H}$ is a real Hilbert space:

Lemma 9. If $\mathcal{H}$ is a Hilbert space over the field $\mathbb{C}$, then if $T : \mathcal{H} \rightarrow \mathcal{H}$ is positive, then T is self-adjoint.

Lemma 10. If $\mathcal{H}$ is a Hilbert space over $\mathbb{R}$, with $T \in \mathcal{B}(\mathcal{H})$, then

$$\|T\| = \sup \{ \langle Tx, y \rangle : \|x\| = 1, \|y\| = 1, x, y \in \mathcal{H} \} .$$

Lemma 11. For $T \in \mathcal{B}(\mathcal{H})$, with T positive and self-adjoint, the map

$$\begin{aligned} S : \mathcal{H} &\rightarrow \mathcal{H} \\ (x, y) &\mapsto (x, Ty) \end{aligned}$$

is sesquilinear, and in particular, Cauchy-Schwarz gives

$$|\langle Tx, y \rangle|^2 \leq |\langle Tx, x \rangle| \cdot |\langle Ty, y \rangle| .$$

T is positive definite if T is injective.

Proof. The map is sesquilinear by the sesquilinearity of the inner product, and because T is a linear operator. Because T is self-adjoint,

$$\begin{aligned} |\langle Tx, y \rangle|^2 &= |\langle Tx, y \rangle| \cdot |\langle x, Ty \rangle| \\ &= |\overline{S(x, y)}| \cdot |S(x, y)| \\ &= |S(x, y)|^2 \\ &\leq S(x, x) \cdot S(y, y) \quad \text{Cauchy-Schwarz in } S \\ &= |\langle Tx, x \rangle| \cdot |\langle Ty, y \rangle| . \end{aligned}$$

Now suppose that T is injective. Let $x \in X$, $x \neq 0$. Because T is injective, $Tx \neq 0$, so we can choose $y \in \mathcal{H}$ so that $\langle Tx, y \rangle \neq 0$. Because

$$|\langle Tx, y \rangle|^2 \leq |\langle Tx, x \rangle| \cdot |\langle Ty, y \rangle| ,$$

$\langle Ty, y \rangle$ cannot be zero, so

$$0 < \frac{|\langle Tx, y \rangle|^2}{|\langle Ty, y \rangle|} \leq |\langle Tx, x \rangle| .$$

So T is positive definite. □

Theorem 27. For $\mathcal{H}$ a Hilbert space, $T \in \mathcal{B}(\mathcal{H})$, and T self-adjoint,

$$\|T\| = \sup_{\|x\|=1} |\langle Tx, x \rangle| .$$

Proof.

□

We use this next result frequently.

Theorem 28. For $\mathcal{H}$ a Hilbert space and $A \in \mathcal{B}(\mathcal{H})$,

$$\overline{\text{range}(A)} = \ker(A^*)^\perp \quad \text{and} \quad \ker(A) = \text{range}(A^*)^\perp.$$

Proof.

- First we prove that $\ker(A) \subset \text{range}(A^*)^\perp$. Let $x \in \ker(A)$ and $y \in \text{range}(A^*)$. Then there exists $z \in \mathcal{H}$ such that $A^*z = y$, and so

$$\langle x, y \rangle = \langle x, A^*z \rangle = \langle Ax, z \rangle = \langle 0, z \rangle = 0.$$

So $x \in \text{range}(A^*)^\perp$.

- Now let $w \in \text{range}(A^*)^\perp$, and let $x \in \mathcal{H}$. Then

$$0 = \langle w, A^*x \rangle = \langle Aw, x \rangle.$$

Since $\langle Aw, x \rangle = 0$ for all $x \in \mathcal{H}$, $Aw \in \mathcal{H}^\perp = \{0\}$, so $w \in \ker(A)$. Thus $\text{range}(A^*)^\perp \subset \ker(A)$.

- We have established that $\ker(A) = \text{range}(A^*)^\perp$. Because $(A^*)^* = A$, we also have

$$\ker(A^*) = \text{range}(A)^\perp,$$

and by taking orthogonal complements,

$$\ker(A^*)^\perp = \text{range}(A) = \overline{\text{range}(A)}.$$

We have equality with the closure because taking the orthogonal complement of a subset of a Hilbert space yields a closed linear subspace. So we have our result.

□

7.2 Three Kinds of Spectral Value

In linear algebra, we talk about a linear map A having eigenvalues and eigenvectors, and they give us a way to characterize the operator by how it scales the eigenspaces. The eigenvalues and eigenvectors are pairs $\lambda \in \mathbb{C}$ and $x \in \mathcal{H}$ such that

$$Ax = \lambda x.$$

We talk about things like the determinant, and sometimes we define the eigenvalues as scalars for which the operator

$$\lambda I - A$$

is not invertible. In finite dimensions, this definition is equivalent to the first; by the rank-nullity theorem, we have the equivalence

$$(A \text{ injective}) \Leftrightarrow (A \text{ surjective}) \Leftrightarrow (A \text{ bijective}).$$

Things get funky in infinite dimensions, because we don't have this equivalence. Linear operators can very well be only injective or only surjective, or neither or both. As a consequence, the equality $Ax = \lambda x$ no longer describes all elements of the spectrum. Instead, we have a handful of definitions.

In the following definitions, A is a bounded linear operator on a Hilbert space, because that's where we will develop most results, but the definitions apply to a broader class of operators.

Definition 14 (point spectrum). *If $\lambda I - A$ is not injective, i.e. if there is $x \neq 0$ such that $Ax = \lambda x$, then we say that λ belongs to the point spectrum, $\sigma_p(A)$ of A . We also call λ an eigenvalue in this case.*

Definition 15 (continuous spectrum). *If $\lambda I - A$ is injective but not surjective, and $\overline{\text{range}(\lambda I - A)} = \mathcal{H}$, then we say that λ belongs to the continuous spectrum, $\sigma_c(A)$ of A .*

Definition 16 (residual spectrum). *If $\lambda I - A$ is injective but not surjective, and $\overline{\text{range}(\lambda I - A)} \neq \mathcal{H}$, then we say that λ belongs to the residual spectrum, $\sigma_r(A)$ of A .*

Definition 17 (resolvent set, resolvent operator). *If $\lambda I - A$ is bijective, then we say that λ belongs to the resolvent set, $\rho(A)$ of A . We call $(\lambda I - A)^{-1}$ the resolvent operator of A at λ .*

Collectively, we refer to the union of the three sets of spectral value as the *spectrum* and denote it by $\sigma(A)$. Going down the list: the point spectrum is the most similar to eigenvalues in finite dimensions, and we will see that the spectral theorem allows us to decompose an operator as it acts on eigenspaces, in the case when the operator is compact and self-adjoint. The elements of the continuous spectrum are almost eigenvalues. Because $\overline{\text{range}(\lambda I - A)} = \mathcal{H}$, i.e. the operator is almost surjective, it can be shown that there is a sequence in $\mathcal{H}$ bounded away from zero that acts in the limit like an eigenvector for λ . Finally, the residual spectrum exists to round out the classification; it is often empty in the examples we consider, and some folks refer to it as the “leftover spectrum”.

The spectrum has applications in physics, for example in the study of unbounded operators, and it has consequences for solving equations. For example, we will later derive the following result:

Theorem 29. *For $T \in \mathcal{B}(\mathcal{H})$ a compact operator, $\sigma(T) \setminus \{0\} \subset \sigma_p(T)$.*

In other words, any nonzero spectral value of T belongs to the point spectrum. This result gives the so-called Fredholm alternative for compact operators:

Corollary 8 (Fredholm alternative). *For $T \in \mathcal{B}(\mathcal{H})$ a compact operator and $\lambda \neq 0$, either*

$$\lambda I - T \text{ is invertible,}$$

or

$$\lambda \in \sigma_p(T).$$

When $\lambda I - T$ defines an equation, as in

$$(\lambda I - T)x = y,$$

the Fredholm alternative tells us any equation of this form satisfies one or the other, but not both:

1. has a unique solution x for any $y \in \mathcal{H}$,
2. has a nonzero solution to $(\lambda I - T)x = 0$.

The most familiar application of this result is in the theory of integral equations. There is some theory devoted to determining when an integral operator is compact. The Fredholm alternative can be used elsewhere, but operations like differentiation, which are common to see in equations, are not compact.

7.3 Topology of the Spectrum

In subsequent sections, we will study the properties of the spectrum that result when an operator meets certain criteria, such as being self-adjoint, compact, or both. In this section, we cover some preliminary results that hold more generally.

Theorem 30. For $(X, \|\cdot\|)$ a Banach space and $A \in \mathcal{B}(X)$,

$$(\|I - A\| < 1) \Rightarrow (A \text{ is invertible}) .$$

Proof.

- To show that A is invertible, let $x \in X$ with $x \neq 0$. Because $\|I - A\| < 1$,

$$\|Ax - x\| < \|x\| .$$

If it were the case that $Ax = 0$, then we would have $\|Ax - x\| = \|0 - x\| = \|x\|$. Based on the above inequality, this is not the case. So $Ax \neq 0$, and A is injective.

- Now we show that A is surjective. With this argument, we actually show injectivity as well. Let $S = I - A$, and define the sequence of operators

$$S_n = \sum_{i=0}^n S^i .$$

We will show that the limit of S_n is A^{-1} .

$$\begin{aligned} AS_n &= (I - S)S_n \\ &= \sum_{i=0}^n S^i - \sum_{i=0}^n S^{i+1} \\ &= S^0 - S^{n+1} \\ &= I - S^{n+1} . \end{aligned}$$

Because $\|S\| < 1$ and we take S to powers,

$$\lim_{n \rightarrow \infty} AS_n = \lim_{n \rightarrow \infty} I - S^{n+1} = I .$$

The same result holds in the other direction:

$$\lim_{n \rightarrow \infty} S_n A = I.$$

We note that because $\|S\| < 1$, the limiting operator $\lim_{n \rightarrow \infty} S_n$ is well defined as a geometric series, so it's our inverse for A .

□

The previous result says that if a bounded operator on a complete space is close enough to the identity, it has an inverse. There is a more general result that says a bounded operator is invertible if it lives in a suitably small neighborhood of another bounded, invertible operator:

Theorem 31. *The set $\text{GL}(X)$ of bounded, invertible linear maps is open in $\mathcal{B}(X)$ for the norm $\|\cdot\|_\infty$.*

We will use the previous result ($\|I - A\| < 1 \Rightarrow (A \text{ invertible})$) to prove:

Theorem 32. *For $(X, \|\cdot\|)$ a Banach space and $A \in \mathcal{B}(X)$, the resolvent set $\rho(A)$ is an open subset of $\mathbb{C}$.*

Proof. Let $A \in \mathcal{B}(X)$, and let $\rho(A)$ be the resolvent set of A . Assume that $\rho(A)$ is nonempty. Let $\lambda_0 \in \rho(A)$, which is to say, $\lambda_0 I - A$ is invertible, and we refer to the corresponding resolvent operator

$$R_{\lambda_0}(A) = (\lambda_0 I - A)^{-1}.$$

Let $\epsilon = [\|R_{\lambda_0}(A)\|]^{-1}$. For $\lambda \in \mathbb{C}$ with $|\lambda - \lambda_0| < \epsilon$, then by scaling the result of theorem 30, we see $\lambda \in \rho(A)$. □

Corollary 9. *The spectrum of $A \in \mathcal{B}(X)$ is compact in $\mathbb{C}$, and*

$$\forall \lambda \in \sigma(A), |\lambda| \leq \|A\|$$

Proof. Because $\rho(A)$ is open, $\sigma(A)$ is closed. By the Heine-Borel theorem, we need only show that $\sigma(A)$ is bounded to get compactness. Let $\lambda \in \mathbb{C}$, $|\lambda| > \|A\|$. Then

$$\|I - \lambda^{-1}(\lambda I - A)\| = \|\lambda^{-1}A\| = |\lambda|^{-1} \cdot \|A\| < 1.$$

From theorem 30, $\lambda^{-1}(\lambda I - A)$ is bijective, and thus $\lambda I - A$ is bijective. So $\lambda \in \rho(A)$. So if $\lambda \in \sigma(A)$, then $|\lambda| \leq \|A\|$. □

So we have a bound on the modulus of spectral values, namely $\|T\|$. It turns out that we can place another bound, called the *spectral radius*.

Theorem 33 (spectral radius). *For $A \in \mathcal{B}(X)$, X a Banach space, the following limit exists, called the spectral radius of A :*

$$r(A) := \lim_{n \rightarrow \infty} \|A^n\|^{1/n},$$

and the following statements hold:

1. $0 \leq r(A) \leq \|A\|$;
2. If $|\lambda| > r(A)$, then $\lambda \in \rho(A)$, and

$$(\lambda I - A)^{-1} = R_\lambda(A) = \sum_{n=1}^{\infty} \lambda^{-n} A^{n-1}$$

with convergence in operator norm.

Corollary 10.

$$\lim_{|\lambda| \rightarrow \infty} R_\lambda(A) = 0 \in \mathcal{B}(X).$$

Proof. From theorem 33,

$$R_\lambda(A) = \sum_{n=1}^{\infty} \lambda^{-n} A^{n-1},$$

so

$$\begin{aligned} \|R_\lambda(A)\| &= |\lambda|^{-1} \cdot \left\| \sum_{n=0}^{\infty} \lambda^{-n} A^n \right\| \\ &\leq |\lambda|^{-1} \cdot \sum_{n=0}^{\infty} \|\lambda^{-n} A^n\| \\ &\leq |\lambda|^{-1} \cdot \sum_{n=0}^{\infty} \left(\|\lambda A\|^{-1} \right)^n. \end{aligned}$$

Because we can consider only λ large enough that $\|\lambda A\|^{-1} < 1$, the geometric series formula gives

$$\|R_\lambda(A)\| \leq |\lambda|^{-1} \cdot \frac{1}{1 - \|\lambda A\|^{-1}}.$$

Because this expression vanishes with large λ ,

$$\lim_{|\lambda| \rightarrow \infty} \|R_\lambda(A)\| = 0.$$

Because X is a Banach space—complete—we have

$$\lim_{|\lambda| \rightarrow \infty} R_\lambda(A) = 0.$$

□

7.4 Self-Adjoint Operators

If $T : \mathcal{H} \rightarrow \mathcal{H}$, is a self-adjoint operator, then for all $x \in \mathcal{H}$, $\langle Tx, x \rangle$ is real. To see why, observe

$$\langle Tx, x \rangle = \langle x, Tx \rangle = \overline{\langle Tx, x \rangle},$$

and because $\langle Tx, x \rangle$ is equal to its complex conjugate, $\langle Tx, x \rangle$ is real. We can say more:

Proposition 2. *Self-adjoint operators have real point spectrum.*

Proof. Let $T : \mathcal{H} \rightarrow \mathcal{H}$ linear, and let $\lambda \in \sigma_p(T)$. Then there is a nonzero $x \in \mathcal{H}$ with

$$Tx = \lambda x.$$

Then

$$\langle Tx, x \rangle = \langle \lambda x, x \rangle = \lambda \|x\|^2.$$

Similarly,

$$\langle Tx, x \rangle = \langle x, Tx \rangle = \langle x, \lambda x \rangle = \bar{\lambda} \|x\|^2.$$

Thus $\lambda \|x\|^2 = \bar{\lambda} \|x\|^2$. Because $x \neq 0$, $\lambda = \bar{\lambda}$. So λ is real. □

It turns out that if T is bounded and self-adjoint, then the whole spectrum is real. To get there, we first establish a lemma:

Lemma 12. *For $\mathcal{H}$ a Hilbert space and $T \in \mathcal{B}(\mathcal{H})$ self-adjoint,*

$$(T \text{ is invertible}) \Leftrightarrow (\exists \gamma > 0, \forall x \in \mathcal{H}, \|Tx\| \geq \gamma \|x\|).$$

Proof.

- Suppose that $T \in \mathcal{B}(\mathcal{H})$ is self-adjoint and invertible. Because T is bijective, T is injective and has closed range. By theorem 11 proved earlier,

$$\exists \gamma > 0, \forall x \in \mathcal{H}, \|Tx\| \geq \gamma \|x\|.$$

- Again, due to theorem 11, if T is coercive, i.e. if the right side of the equivalence holds, then T is injective with closed range. So we will prove that T is also surjective.

Because $\ker(T) = 0$, and because T is self-adjoint, $\ker(T^*) = 0$. From theorem 28, and because T is self-adjoint,

$$\overline{\text{range}(T)} = \ker(T^*)^\perp = \ker(T)^\perp.$$

Because $\text{range}(T)$ is closed,

$$\text{range}(T) = \ker(T)^\perp = \{0\}^\perp = \mathcal{H}.$$

So T is surjective.

□

Corollary 11. *If $A \in \mathcal{B}(\mathcal{H})$ is self-adjoint, then for*

$$m := \inf_{\|x\|=1} \langle Ax, x \rangle,$$

$$M := \sup_{\|x\|=1} \langle Ax, x \rangle,$$

the spectrum is bounded according to $\sigma(A) \subset [m, M]$.

Proof. Because A is self-adjoint, $\sigma(A)$ is real. So we check that $\sigma(A)$ is bounded in the interval $[m, M]$.

Let $\lambda < m$. We will show that $\lambda \notin \sigma(A)$. Let $x \in \mathcal{H}$, $\|x\| = 1$. Then

$$\langle (A - \lambda I)x, x \rangle = \langle Ax, x \rangle - \lambda \geq m - \lambda =: \epsilon > 0.$$

By Cauchy-Schwarz,

$$\begin{aligned} |\langle (A - \lambda I)x, x \rangle| &\leq \|(A - \lambda I)x\| \cdot \|x\| \\ &= \|(A - \lambda I)x\|, \end{aligned}$$

so $\|(A - \lambda I)x\| \geq \epsilon$, and for any $u \in \mathcal{H}$,

$$\|(A - \lambda I)u\| \geq \epsilon\|u\|.$$

We have shown that $A - \lambda I$ is coercive, and because $A - \lambda I$ is self-adjoint, $A - \lambda I$ is bijective. So $\lambda \notin \sigma(A)$.

The proof of the other direction uses the same argument to show that if $\lambda > M$, then $\lambda I - A$ is bijective. $\square$

8 Measure and Integration

8.1 Limit Theorems for Lebesgue Integration

Theorem 34 (monotone convergence theorem). *Let (X, Σ, μ) be a measure space, and let $\{f_n\}_{n \in \mathbb{N}}$ be a sequence of measurable functions such that*

$$0 \leq f_1(x) \leq f_2(x) \leq \dots \quad \forall x \in X,$$

and

$$\lim_{n \rightarrow \infty} f_n(x) = f(x) \quad \forall x \in X.$$

Then f is measurable, and

$$\lim_{n \rightarrow \infty} \int_X f_n d\mu = \int_X f d\mu.$$

The following lemma is useful because of how loose the requirements are for its application—the $\liminf$ always exists. The proof is from [?].

Lemma 13 (Fatou's lemma). *For (X, Σ, μ) a measure space and $\{f_n\}_{n \in \mathbb{N}}$, $f_n : X \rightarrow [0, \infty]$ a sequence of measurable functions, then*

$$\int_X \left(\liminf_{n \rightarrow \infty} f_n \right) d\mu \leq \liminf_{n \rightarrow \infty} \int_X f_n d\mu.$$

Proof. Let

$$g_k(x) = \inf_{i \geq k} f_i(x) \quad \forall k \in \{1, 2, 3, \dots\}, \quad \forall x \in X.$$

First, we can see that for any k , $g_k \leq f_k$, and so

$$\int_X g_k d\mu \leq \int_X f_k d\mu.$$

By definition, $\lim_{k \rightarrow \infty} g_k = \liminf_{n \rightarrow \infty} f_n$. We also see that $g_k \leq g_{k+1}$ for all k . So we can use the above inequality and apply the monotone convergence theorem on the left hand side as follows:

$$\begin{aligned} \int_X g_k d\mu &\leq \int_X f_k d\mu \\ \liminf_{k \rightarrow \infty} \int_X g_k d\mu &\leq \liminf_{k \rightarrow \infty} \int_X f_k d\mu \\ \lim_{k \rightarrow \infty} \int_X g_k d\mu &\leq \liminf_{k \rightarrow \infty} \int_X f_k d\mu \\ \int_X \left(\lim_{k \rightarrow \infty} g_k \right) d\mu &\leq \liminf_{n \rightarrow \infty} \int_X f_n d\mu \\ \int_X \left(\liminf_{n \rightarrow \infty} f_n \right) d\mu &\leq \liminf_{n \rightarrow \infty} \int_X f_n d\mu. \end{aligned}$$

□

You can't always get away with swapping a limit and integral. In the usual counterexamples of integration theory, the sequence of functions has some limiting behavior with a narrow pocket of density escaping toward infinity. The dominated convergence theorem formalizes this idea. If your sequence of functions is pointwise bounded by some integrable function, then limits behave as expected—no funny business. It reminds of the uniform boundedness principle.

Theorem 35 (dominated convergence theorem). *Let (X, Σ, μ) be a measure space and $\{f_n\}_{n \in \mathbb{N}}$ a sequence of measurable functions such that*

$$f(x) = \lim_{n \rightarrow \infty} f_n(x) \text{ exists for all } x \in X,$$

and suppose that there is an integrable function g (i.e., $\int_X |g| d\mu$ exists and is finite), such that $|f_n(x)| \leq g(x)$ always. Then

$$\lim_{n \rightarrow \infty} \int_X |f_n - f| d\mu = 0,$$

and

$$\lim_{n \rightarrow \infty} \int_X f_n d\mu = \int_X f d\mu.$$

Proof. Because $|f| \leq g$ and f is measurable, f is integrable. By triangle inequality, $|f_n - f| \leq 2g$, so we can use Fatou's lemma with the function $2g - |f_n - f|$. By Fatou's lemma,

$$\begin{aligned} \int_X 2g d\mu &\leq \liminf_{n \rightarrow \infty} \int_X (2g - |f_n - f|) d\mu \\ &= \int_X 2g d\mu + \liminf_{n \rightarrow \infty} \int_X (-|f_n - f|) d\mu \\ &= \int_X 2g d\mu - \limsup_{n \rightarrow \infty} \int_X |f_n - f| d\mu. \end{aligned}$$

We know that $\int_X 2g d\mu$ is finite, so we're safe subtracting it away to obtain

$$\limsup_{n \rightarrow \infty} \int_X |f_n - f| d\mu \leq 0,$$

and we conclude that $\lim_{n \rightarrow \infty} \int_X |f_n - f| d\mu = 0$. Because

$$\left| \int_X f_n - f d\mu \right| \leq \int_X |f_n - f| d\mu,$$

we get

$$\lim_{n \rightarrow \infty} \left| \int_X f_n d\mu - \int_X f d\mu \right| = 0,$$

so

$$\lim_{n \rightarrow \infty} \int_X f_n d\mu = \int_X f d\mu.$$

□